

A Multi-Agent Smart-City Simulation for Flow-Responsive Traffic and Energy Management with Decision-Tree Optimization

Matthew Mitchell

Computer Science M.S. - Artificial Intelligence Certificate
Full Sail University, Winter Park, FL, USA

Abstract- This project implements a reproducible smart-city simulation that coordinates traffic-light behavior and zone-level energy usage across four intersections and four energy zones. Four local traffic-control agents use rule-based queue observations rather than fixed-cycle timing. A 16-configuration parameter sweep supplies training data to a DecisionTreeRegressor that predicts average traffic delay and identifies a low-delay timing configuration. Under a matched random seed, the optimized timing reduced average wait from 4.69 s to 4.09 s, a 12.9% reduction. A separate 24-hour energy model applies time-of-day demand multipliers and automated 15% load shedding above a 160 kW zone threshold. After correcting energy aggregation to respect the one-hour sample interval, the model delivered 9.286 MWh of 9.518 MWh requested, shedding 0.232 MWh and producing a 97.6% demand-served ratio. Static plots, tabulated metrics, logged JSON artifacts, and a read-only Unreal Engine replay layer provide evidence of system behavior without duplicating the simulation logic in the visualization environment.

Index Terms- smart city, multi-agent simulation, traffic signal control, discrete-event simulation, decision tree regression, energy management, Unreal Engine

I. INTRODUCTION

Urban infrastructure is a coupled decision problem. Traffic demand varies by location and time, while energy demand varies across buildings and zones. Fixed traffic schedules and unconstrained peak-energy delivery can be adequate under stable conditions, but both approaches become less effective when demand changes faster than the control policy. Contemporary smart-city research therefore treats traffic intersections, buildings, and other infrastructure elements as autonomous or semi-autonomous agents that respond to local state while contributing to a network-level objective [1], [4].

This project addresses the assignment as an integrated but deliberately modular simulation. The traffic subsystem is a discrete-event model implemented with SimPy. Four intersections operate as local rule-based control agents that observe simulated queue state and determine phase duration. The energy subsystem independently samples four demand zones over a 24-hour day and applies a threshold-based load-shedding policy. A scikit-learn decision-tree regression model is trained on the outcomes of a systematic traffic-light parameter sweep and is used to identify a low-wait timing configuration. pandas and NumPy compute performance metrics, Matplotlib produces analysis figures, and Unreal Engine is used as a read-only replay environment for the logged simulation state.

The design intentionally separates simulation truth from presentation. Traffic events, light-state transitions, energy records, optimization results, and summary metrics are produced by the Python pipeline first. Visualization software consumes those artifacts but does not create substitute traffic outcomes. This separation makes the result auditable: screenshots and video can be traced to the same event log used for quantitative analysis. It also addresses the course AI-use policy because generated or AI-assisted code is not treated as evidence by itself; the reported claims are tied to

executable outputs, seeded comparisons, and dimensional checks.

II. RELATED WORK

Recent work demonstrates why traffic signal control is naturally framed as a multi-agent problem. Olusanya et al. represent traffic signals as agents in a SUMO environment and evaluate multi-agent reinforcement learning using waiting time, queue length, and congestion metrics [1]. Poudel et al. extend adaptive signal control to jointly consider pedestrian and vehicle delay along an eight-signal urban corridor, using real demand observations to train and evaluate a deep-RL controller [2]. These studies motivate local, state-responsive intersection control even though the present project uses explainable rules rather than learned policies for moment-to-moment light changes.

Almomany et al. compare fixed-time, pressure-based, delay-based, hybrid, and reinforcement-learning traffic-control methods using microscopic traffic simulation in a Kuwait City application context [3]. Their work is useful here as a methodological benchmark rather than as evidence of a physical city deployment. On the energy side, Zhang et al. present STEMS, a safety-constrained multi-agent reinforcement-learning framework for coordinated building-energy management that explicitly models spatial and temporal relationships among buildings [4]. The present energy controller is simpler, but it follows the same general principle that distributed demand should be observed and managed rather than assumed constant.

Nuthakki demonstrates the value of pairing an analytical traffic model with a three-dimensional engine-based simulation, using Unity and PPO to study safety, traffic flow, fuel efficiency, and emissions [5]. That work supports the use of Unreal Engine here as an evidence and replay layer. The distinction is important: this project does not claim that Unreal is the simulation engine or that its visual agents learn.

The Python model remains authoritative, while Unreal visualizes logged state so that the presentation layer cannot silently diverge from the measured experiment.

III. SYSTEM DESIGN

A. City Environment and Agent Model

The simulated city contains four intersections, labeled I-NW, I-NE, I-SW, and I-SE, arranged around a road ring. Four energy zones occupy the same city-scale model. Each intersection has north-south and east-west approaches and an independent traffic-light controller. In the multi-agent interpretation used for this assignment, each controller is a local traffic agent: it owns its light state, receives queue observations, and makes phase-extension decisions without a single global fixed-time schedule. The agents share the same traffic-generation assumptions but maintain independent local state.

Vehicle arrivals follow exponential inter-arrival times corresponding to a Poisson process with a mean rate of six vehicles per minute per simulated approach. SimPy processes create arrivals, queues, controller ticks, releases, and timestamps. The controller loop advances at 0.5-s resolution, while vehicle service is limited to one queued vehicle per 2.5 s of green on an eligible axis. Arrival time, release time, wait time, vehicle type, intersection identity, and light state are written to the logged output used by later analysis and visualization.

B. Flow-Responsive Traffic-Light Automation

Each traffic agent uses a bounded gap-out/max-out policy instead of a fixed-cycle timer. A green phase is guaranteed for `min_green` seconds. After that minimum, queue-state conditions determine whether the active phase remains green for another control increment, subject to the hard `max_green` ceiling and a fixed yellow transition. This rule structure responds to simulated demand while retaining deterministic safety bounds. It therefore satisfies the assignment requirement for traffic-light automation without overstating the implementation as reinforcement learning.

C. Energy-Grid Management

The energy subsystem models four zones with distinct base loads. Each zone is sampled once per hour for 24 hours. A time-of-day multiplier raises demand to 1.15 times baseline from 07:00 through 21:00 and reduces demand to 0.45 times baseline overnight. Multiplicative Gaussian variation with mean 1.0 and standard deviation 0.08 prevents the demand profile from becoming unrealistically deterministic. If a zone-hour raw demand exceeds 160 kW, the controller sheds 15% of that zone's entire raw demand for that hour. The model therefore generates both requested demand and delivered load, permitting direct measurement of energy curtailed by automation.

D. Decision-Tree Optimization

Two traffic parameters are exposed to optimization: `min_green` in {3, 5, 7, 10} s and `max_green` in {15, 20, 25, 30} s. All 16 valid combinations are simulated for 900 s under the same sweep seed. For each trial, the pipeline stores average wait, a simple idling-energy proxy, and the number of vehicles served. A DecisionTreeRegressor with maximum depth 4 is fit to the 16 parameter-outcome observations. The

predictive target is average wait time; the selected configuration is the model-supported candidate with the minimum delay among the tested parameter set.

The idling-energy proxy is intentionally not treated as a calibrated physical fuel model. It is a monotonic penalty derived from aggregate waiting time, so reducing waiting also reduces the proxy within the sweep. Building-energy demand remains an independent physical-unit calculation. Keeping the two quantities separate prevents the analysis from combining an illustrative traffic penalty with the zone-level electricity model.

E. Unreal Engine Replay and Evidence

The final visualization deliverable uses Unreal Engine as a replay client rather than as a second simulator. The Python pipeline exports light-state transitions and vehicle-release events with simulation timestamps. Unreal reads those artifacts, places the corresponding vehicles and signals in a city scene, and advances visual state from the recorded timestamps. It does not recompute queues, select light timings, or generate replacement metrics. Figure 1 summarizes this architecture. The companion evidence package includes simulation screenshots, code-execution screenshots, generated graphs, and a short Unreal recording so that both quantitative output and visible system behavior can be inspected.

IV. EXPERIMENTAL METHOD AND VALIDATION

Three traffic runs are kept separate to avoid training-test ambiguity. First, the 16-point sweep uses one fixed seed and 900 simulated seconds per timing pair; these outcomes train the regression tree. Second, the default configuration (`min_green` = 5 s, `max_green` = 25 s) and selected configuration (3 s, 20 s) are each rerun for 900 s using a second shared seed. This matched-seed comparison isolates the timing-policy difference from random arrival variation. Third, the selected configuration is run for 1800 s under a third seed to generate a longer event log and the reported Task 6 throughput statistics. The energy model independently covers 24 hours x 4 zones, producing 96 zone-hour records.

The principal traffic improvement is calculated from the unrounded matched-run values. Let W_d denote the default average wait and W_o the optimized average wait. The percentage reduction is

$$\text{Improvement} = ((W_d - W_o) / W_d) \times 100 \quad (1)$$

Energy aggregation was validated dimensionally because each record represents one full hour. Let $D_{z,t}$ be requested power, $L_{z,t}$ delivered power, and $\Delta t = 1$ h. Total shed energy is the sum of $(D_{z,t} - L_{z,t})\Delta t$ across zones and hours; no division by 60 is appropriate for hourly samples. The demand-served ratio is

$$\text{Demand served} = (\Sigma E_{del} / \Sigma E_{dem}) \times 100 \quad (2)$$

A unit-validation check confirmed that hourly samples must be integrated with $\Delta t = 1$ h; applying an unnecessary minute-to-hour conversion would understate shed energy by a factor of 60. The verified aggregate is 231.65 kWh shed. This check demonstrates verification of AI-assisted computation rather than acceptance of generated output without unit analysis.

V. RESULTS

A. Traffic Performance

The decision tree fits the 16 sweep observations with $R^2 = 1.000$ and selects `min_green = 3 s` and `max_green = 20 s`. Because R^2 is calculated on the same small data set used for fitting, it is reported only as training fit and not as evidence of out-of-sample generalization. In the matched-seed comparison, default average wait is 4.6897 s and optimized average wait is 4.0856 s. Equation (1) therefore gives a 12.88% reduction, reported as 12.9%. Figure 2 shows the comparison.

The longer 1800-s optimized run produces an average wait of 3.97 s, a 95th-percentile wait of 8.89 s, and throughput of 46.48 vehicles per minute. The sweep also indicates that lower delay is not achieved simply by serving fewer vehicles: the 16 trials serve between 707 and 711 vehicles in 900 s, a spread of only four vehicles. Within the sweep, the default pair (5,25) has an idling proxy of 63.90, while the selected pair (3,20) has 54.93, a 14.0% reduction in that stated proxy.

The parameter landscape provides an additional sensitivity result. Averaged over `max_green` values, mean wait rises from 3.88 s at `min_green = 3 s` to 4.51 s at 5 s, 5.38 s at 7 s, and 5.89 s at 10 s. By contrast, averaging over `min_green` values changes mean wait only from 4.95 s at `max_green = 15 s` to approximately 4.90 s at 25-30 s. In this small experiment, minimum-green duration therefore exerts the stronger effect on delay. Several `min_green = 3 s` configurations are nearly tied at displayed precision, so the selected maximum-green value should not be interpreted as a universal optimum outside the tested range.

B. Energy Performance

The energy model produces a mean delivered load of 96.73 kW per zone-hour and a peak observed delivered zone load of 158.57 kW. Across the 96 zone-hours, delivered energy is 9,286.03 kWh and requested energy is approximately 9,517.69 kWh. The controller therefore sheds 231.65 kWh (0.232 MWh) during threshold events and serves 97.57% of requested energy. The term demand-served ratio is used instead of energy efficiency because the metric describes how much requested energy was delivered, not conversion efficiency. Figure 3 shows the traffic-density and zone/hour energy patterns.

The results show two distinct automation effects: traffic control reduces delay through responsive phase timing, while energy control caps high-demand zone-hours through curtailment. They are not combined into a single optimization score. This separation preserves interpretability and makes it possible to improve either subsystem later without changing the other subsystem's reported units or objective.

C. Reproducibility and Evidence

Table I reports every timing combination used to train the decision tree. Figure 4 visualizes the same sweep as a response surface. The numerical paper results are generated from stored artifacts rather than manually estimated from the plots. The final submission package pairs those artifacts with the source code and Unreal replay evidence. This creates a traceable chain from simulation inputs, to logged state, to pandas metrics, to figures, to the engine visualization. Such traceability is especially important when AI-assisted coding

tools are used because it distinguishes executable evidence from plausible-looking generated code.

VI. DISCUSSION AND LIMITATIONS

The experiment demonstrates the required concepts, but its scope remains intentionally small. The decision tree has only two input features and 16 training points; a perfect training fit is expected to overstate predictive certainty. A stronger study would use denser timing ranges, repeated stochastic seeds, a held-out validation set, and either cross-validation or a second simulation scenario. The current result should therefore be interpreted as optimization within a bounded candidate grid, not as a learned controller that generalizes to arbitrary cities.

The traffic agents are rule-based, not reinforcement-learning agents. Their value is explainability: phase duration can be traced directly to queue-state rules and timing bounds. The trade-off is that they do not learn long-horizon coordination policies. Similarly, the building-energy controller uses a fixed threshold and fixed shedding fraction rather than forecast-aware or multi-objective control. Future work could replace either controller with RL while retaining the same logging, validation, and Unreal replay interfaces.

The largest current modeling limitation is realism rather than software architecture. The road network contains four intersections, arrivals use a single statistical family, and the energy model does not include prices, renewable generation, storage, occupant comfort, or feeder constraints. The Unreal environment improves observability and evidence quality but does not by itself increase simulation fidelity. Fidelity must come from better models, calibrated parameters, and stronger validation data.

VII. CONCLUSION

This project satisfies the smart-city simulation objectives through a four-agent traffic environment, four-zone energy model, responsive traffic-light rules, automated load shedding, a decision-tree regression optimizer, quantitative performance analysis, and an Unreal Engine replay layer. The optimized light timing reduces matched-seed average wait by 12.9% and reduces the stated idling proxy by 14.0% within the sweep, while the longer optimized run sustains 46.48 vehicles/min with a 3.97-s mean wait. The corrected energy calculation delivers 9.286 MWh of 9.518 MWh requested and sheds 0.232 MWh, yielding a 97.6% demand-served ratio. Most importantly, the project separates simulation, optimization, analysis, and visualization so that each reported result can be checked against logged evidence rather than inferred from presentation alone.

ACKNOWLEDGMENT

AI-assisted development and review tools, including Claude and ChatGPT, were used to assist with coding, analysis, and document refinement in accordance with the course policy. All reported performance values were checked against executable simulation artifacts, and the energy aggregation was independently validated by dimensional analysis before final reporting.

RESULTS FIGURES AND TABLE

Simulation-to-Visualization Architecture

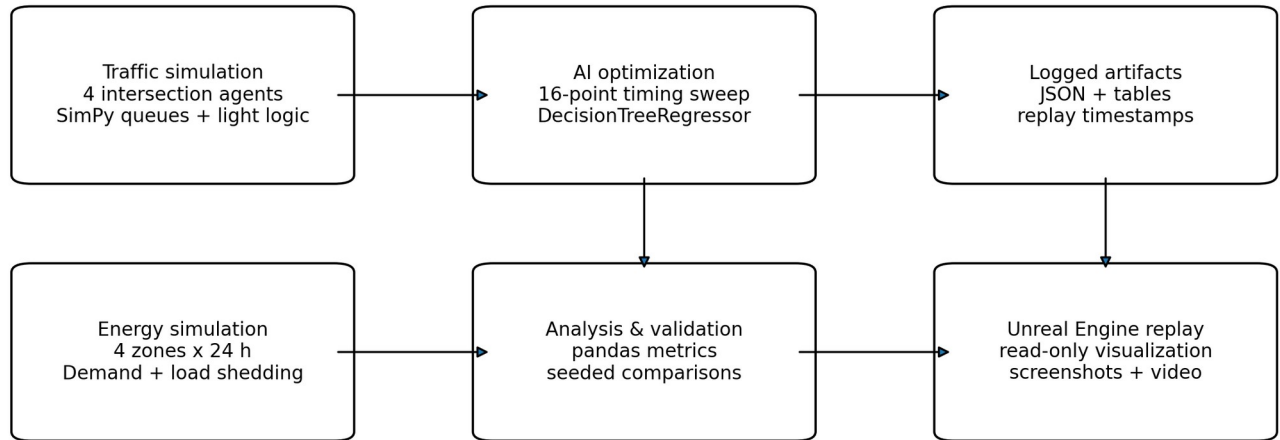


Fig. 1. Simulation-to-visualization architecture. Unreal Engine consumes logged state as a replay client; it does not recompute the simulation.

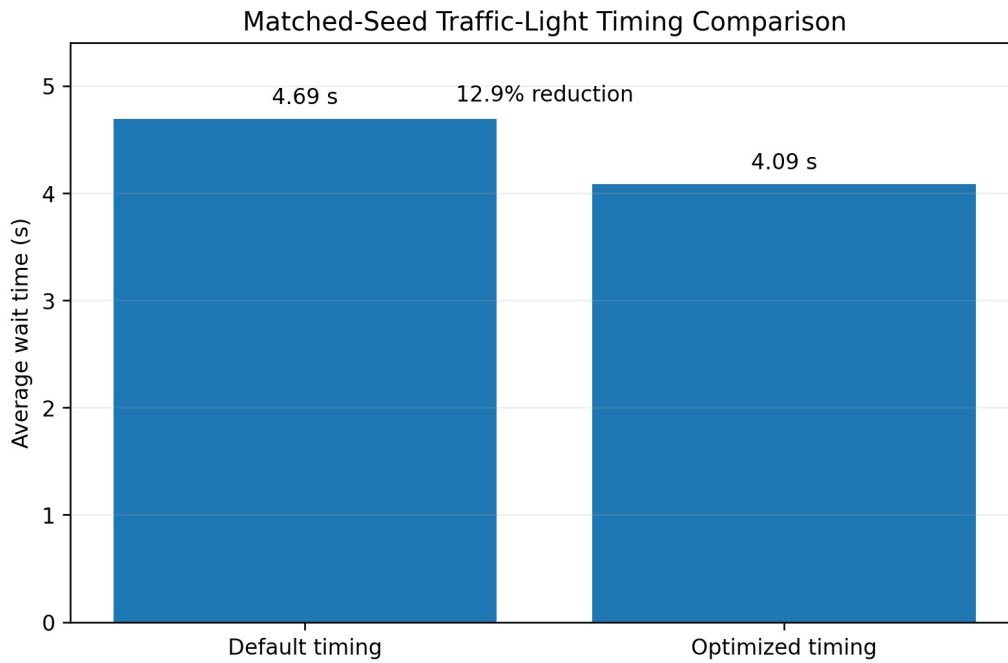


Fig. 2. Matched-seed comparison of default and optimized timing. Average wait decreases from 4.69 s to 4.09 s (12.9%).

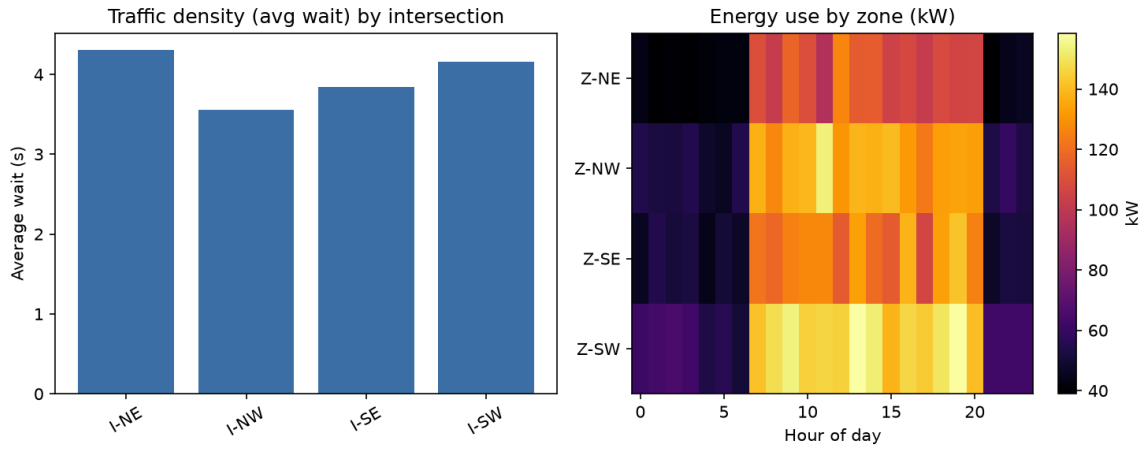


Fig. 3. Per-intersection average traffic wait and delivered energy load by zone and hour. The energy heatmap shows the expected day/night demand pattern.

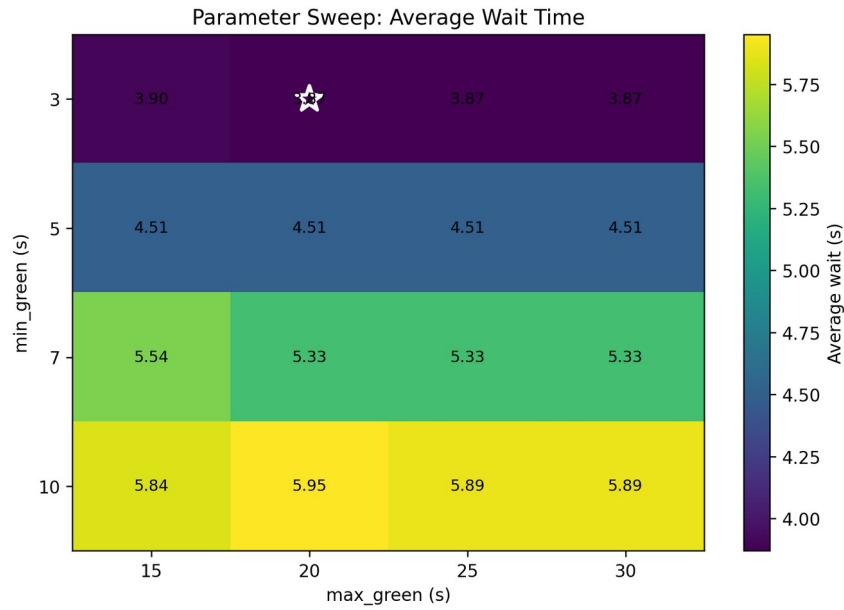


Fig. 4. Average wait across the full 16-point timing sweep. The star marks the selected (min_green = 3 s, max_green = 20 s) configuration.

TABLE I
PARAMETER SWEEP RESULTS (900 s PER TRIAL)

min_green (s)	max_green (s)	avg. wait (s)	idle proxy (source units)	vehicles served (900 s)
3.0	15.0	3.90	55.37	709
3.0	20.0	3.87	54.93	709
3.0	25.0	3.87	54.93	709
3.0	30.0	3.87	54.93	709
5.0	15.0	4.51	63.82	708
5.0	20.0	4.51	63.90	709
5.0	25.0	4.51	63.90	709
5.0	30.0	4.51	63.90	709
7.0	15.0	5.54	78.83	711
7.0	20.0	5.33	75.70	710
7.0	25.0	5.33	75.70	710
7.0	30.0	5.33	75.70	710
10.0	15.0	5.84	82.60	707
10.0	20.0	5.95	84.18	708
10.0	25.0	5.89	83.40	708
10.0	30.0	5.89	83.40	708

Note: The idle quantity is a stated monotonic traffic penalty from the source output; it is not the building-energy model and is not used as a calibrated physical fuel estimate.

REFERENCES

- [1] O. O. Olusanya, Y. Owosho, I. Daniyan, A. W. Elegbede, Q. B. Sodipo, A. Adeodu, H. S. Phuluwa, T. K. Ramasu, and M. G. Kana-Kana Katumba, "Multi-agent reinforcement learning framework for autonomous traffic signal control in smart cities," *Frontiers in Mechanical Engineering*, vol. 11, art. 1650918, Sep. 2025, doi: 10.3389/fmech.2025.1650918.
- [2] B. Poudel, X. Wang, W. Li, L. Zhu, and K. Heaslip, "Joint pedestrian and vehicle traffic optimization in urban environments using reinforcement learning," *arXiv:2504.05018*, 2025.
- [3] A. Almomany, E. Eedi, and M. Sutcu, "Real-time traffic signal optimization for urban mobility: A reinforcement learning-enhanced framework with application to Kuwait City," *Frontiers in Robotics and AI*, vol. 12, art. 1669952, Sep. 2025, doi: 10.3389/frobt.2025.1669952.
- [4] H. Zhang, D. Wu, A. Zinflou, and B. Boulet, "STEMS: Spatial-temporal-enhanced safe multiagent coordination for building energy management," *IEEE Internet of Things Journal*, vol. 13, no. 5, pp. 9548-9560, Mar. 2026, doi: 10.1109/JIOT.2025.3645808.
- [5] M. Nuthakki, "Safety-prioritized, reinforcement learning-enabled traffic flow optimization in a 3D city-wide simulation environment," *arXiv:2506.03161*, 2025.
- [6] A. M. Law, *Simulation Modeling and Analysis*, 5th ed. New York, NY, USA: McGraw-Hill, 2015.